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Innovation:
A Customer-Driven Approach

Introduction

No well-intentioned company would try to develop a product that did not succeed in the marketplace, but stories of product failures in fact abound. Sometimes the product or service may be sound but marketing and distribution issues destroy its chances for success. In other cases, the product or service itself may not meet users' latent and stated requirements. What can be done during the development process to assure that the output successfully exhibits the qualities that customers want to see? Put in simple terms, if a company uncovers what customers want, designs a product that meets those needs, and is able to build and distribute it at an acceptable cost, the company will succeed – although there are innumerable ways in which that simple formula can fail. This note will address the issues of product development broadly and suggest how to improve the probability of a project's success.

Laying the Foundation

The Innovation Funnel

A helpful way of thinking about innovation is to view the process as a funnel (see **Figure 1**). The widest opening, indicating the point of maximum flexibility, exists at the beginning of the project. As the project evolves, the company commits to specific concepts, product designs, and manufacturing processes. This reduces the options available to the product developers, i.e., the funnel narrows.

Before "entering" the project funnel, company management needs to address questions of the project's scope (e.g., basic project objectives, profitability, introduction date) and the resources it will likely consume. More often than not, companies have more projects underway than they can possibly complete with the available resources in the desired time frame. As a result, no project should be approved until the entire portfolio of projects has been considered and the real feasibility of completing all these efforts has been assessed. And when companies face rapidly changing

Research Associate Ellen Stein prepared this note under the supervision of Professor Marco Iansiti and Visiting Professor Thomas J. Kosnik as the basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

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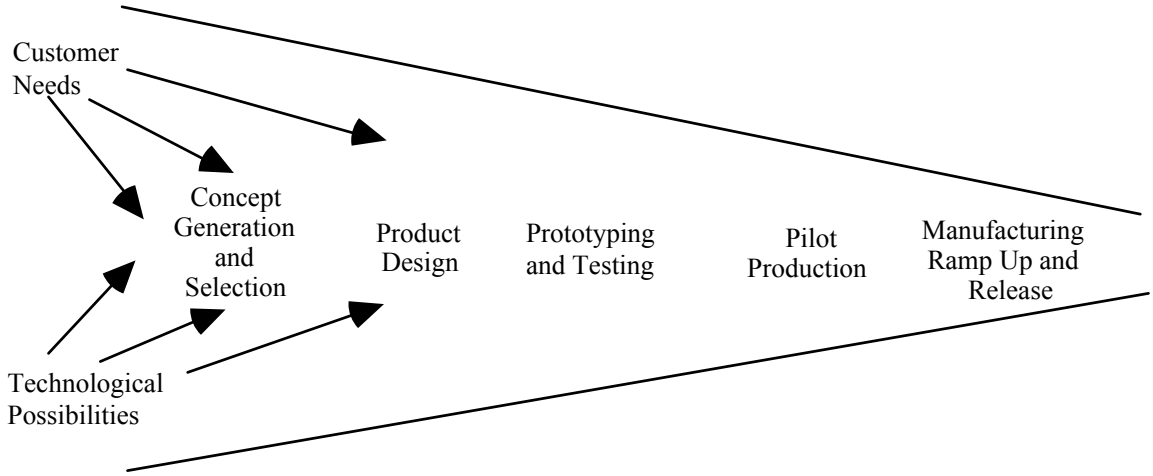
market conditions, projects initially approved to enter the innovation funnel may subsequently have to be abandoned, based on market intelligence gained later in the development process.

Figure 1: The Innovation Funnel



A Complementary View: The Stage-Gate Model

Another view of the new innovation process used in some organizations is a Stage-Gate Model. This conceptualization of the process identifies a series of development stages, similar to those shown in the innovation funnel. It complements the funnel by explicitly identifying a series of gates through which a project must successfully pass on its journey from initial idea to full commercial launch.

Each gate can be a go/no-go decision. At each gate, the project can also be compared with other projects in the company's innovation funnel that are competing for scarce development resources. A company often must decide to put some projects on the "back burner," or even to cancel projects, based on information about changing customer needs, technology, or competitive activity.

A Stage-Gate Model creates discipline in the process by requiring a periodic systematic review of every project in the pipeline at multiple milestones in its development cycle. This allows management of a company to "pull the plug" on new product projects that may have seemed like a good idea at the beginning, but that are less attractive as new information becomes available. By periodically "thinning the herd," a company increases the chances for success of projects that remain in the funnel, by ensuring that they will receive the resources they need to move rapidly through the development process to market launch.

Figure 2 is a schematic of a Stage-Gate Model, as developed by Cooper.[1] It is important to note that although the Stage-Gate Model appears to be a linear sequence of activities, most companies with effective innovation processes allow for cross functional teams to work on parallel, overlapping tasks throughout a project. For example, a project team may develop tentative plans for testing and validation (stage 4) and full production (stage 5) before a project has passed through Gate

[1] See Cooper, "Stage-Gate Systems: A New Tool For Managing New Products".

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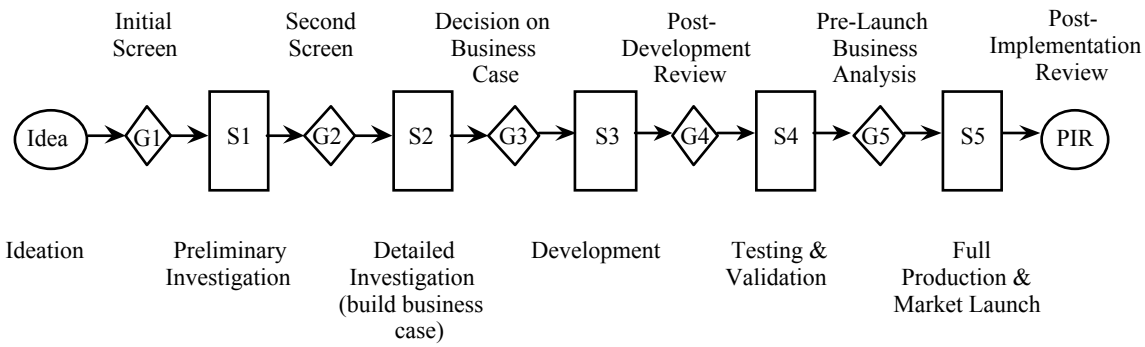
3 or Gate 4. However, if a product fails at one of those gates, the plans for later stages are not implemented.

Figure 2: The Stage-Gate Model of Innovation



Why are the funnel and the Stage-Gate Model important? First, because they offer complementary tools to help guide new team members through the seemingly chaotic process of their first innovation process. Second, because they allow management of a company to manage the many projects in their new product or service portfolio in a dynamically changing environment, and to make decisions that optimize the allocation of resources across projects as markets evolve. Third, because they create expectations among those that work in the new product development process that not every good idea will necessarily reach the market. This may reduce the disappointment among project team members who invest their time and energy in a project, only to have it postponed or canceled. Finally, these tools can be used to diagnose potential problems in the product development process. For example, an innovation funnel with too wide a mouth might symbolize a process where too many ideas are allowed to start the process. A Stage-Gate Model with a gate that only 1% of the projects survive might indicate a screening process that is so risk-averse that it is excessively constrains a company's capacity for innovation.

Assessing Market Potential

At the outset of an innovation project, a preliminary assessment is usually made of the potential demand for the new product or service. The estimate of market potential is revised and refined with the most current information about costs, competition, technology, and market needs at every gate of the Stage-Gate Model. The assessment of market potential is closely linked to the selection of the market segment that a new product will serve. Why is the choice of target market so important? A CEO of a leading software company (who wishes to remain anonymous) put it best:

> It usually takes about the same amount of development time and talent to design a product that will meet the needs of one hundred customers as a product that will meet the needs of one million. Since talented product developers are usually a scarce resource, most shareholders, customers, and employees would prefer that a company leverage that resource to meet the greatest need. For any business that aspires to grow and prosper, picking the right market segment is the most critical decision in the product development process.

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There are numerous tools and techniques for estimating size and growth potential of a product category, or the market share of a new entrant that aspires to challenge existing competitors.

Most make creative use of secondary data that is relatively inexpensive to acquire, and is often free via libraries, government agencies, and so forth. A few basic approaches for estimating market demand are shown below:

- Population-based estimates: These estimates involve taking the total population, number of households, number of companies, and so forth, and estimating the percentage that will buy in a product category each year. This technique is often used for consumer durables such as color televisions, cellular phones, or automobiles. For a new product category, there are numerous models that predict the number of units bought over time will resemble an S-shaped curve, whose slope depends on the percentage of the population that are "innovators" and "imitators" (who buy partially because they saw that someone else had purchased), and the saturation point (the total percentage of the population that will buy over the product category's life cycle).

- Population * usage-based estimates: For product categories that have a short interval between purchases, such as cola drinks, coffee, and toilet paper, the market size in any year is a function of the number of customers (individuals or households) multiplied by the number of purchase occasions per year, and the average number of units purchased on each occasion.

- Forecasting by analogy: For new product categories without an existing track record, examining the market size of similar products in the first few years of their life cycle offers a baseline for the new product. For example, a company estimating the market for a new movies-on-demand service via the information superhighway might examine the early growth records of cable movie channel subscriptions, such as Showtime, Cinemax, and Home Box Office, to determine how many households might become customers of movies-on-demand each year for the first several years.

- Forecasting by percentage of complementary product sales: For example, the creators of a new software program for notebook computers might assume that they could achieve a percent penetration of all new notebook computers sold, and take advantage of secondary research data on the number of notebook computers that will be sold in the next few years. Similarly, the number of bicycle helmets might be tied to bicycle sales.

- Forecasting based on past competitor performance: In product categories where other companies have established a track record, estimating a new entrant's share of the market can be based partially on the experience of competitors. For example, if only one in ten new products in a category has been successful, and if there are data available about the investment that various competitors have made in trying to launch new products, as well as first year unit sales for different launches, a company can build a pragmatic estimate of the range of sales that a new product might enjoy in best case, expected, and worst case scenarios.

The five approaches above suggest that a quick-and dirty estimate of market potential for a new product or service category and for a new brand in an existing category can be done with a

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minimal investment of time and money. The irony is that a task that is relatively straightforward is often wildly inaccurate. Many prognostications have erred on the side of over-optimism; consider, for example, the forecasts of demand for eight track tape systems in the 1960s, or artificial intelligence software in the 1980s.

What lessons can we learn from all of this? First, estimating market potential is important. Second, we should be willing to challenge our own forecasts, or those of our suppliers and customers. Third, we should plan for contingencies so that if the demand for our new product is ten times less or ten times more that we had anticipated, we have a game plan to respond to the voice of the marketplace as it makes itself heard during the new product launch. Finally, we should reduce cycle time in new product development, manufacturing, and delivery so that we can develop and deliver products we know customers will buy – because they've already ordered them in advance.

Selecting and Managing the Development Team

Once the innovation process is understood, the formation and operation of the development team are critical to the project's outcome. Numerous factors influence optimal team composition, including company size, reporting structure, geographical proximity of employees, product nature (e.g., technical, marketing-driven), available resources, and so forth. In most environments small, cross-functional teams are recommended if possible.[2] In addition, it is important to create incentives for the team members to work collaboratively, and infrastructure to help the team communicate and work effectively across barriers of time, distance, and cultural differences that might otherwise inhibit its effectiveness.

Developing the Product

Once the team has been selected and the project scope defined, actual product/service development can begin. By this point, general issues of target markets and financials have been addressed in the project scope definition, but no specifics about the product itself should have been outlined. That discussion is initiated through the concept-generation process.

Preparing for Concept Generation

A product concept addresses the question of how the product will meet stakeholder needs by describing, usually through a drawing or rough three-dimensional model and some general text, what technology and form the product will employ. Before that sketch or description can be created, however, the needs that the product or service will fulfill have to be uncovered. For this to happen, a decision must be made about whose needs will be met.

Who Are the Stakeholders?

Quite often, companies think of getting input only from "the customers," i.e., from those who either make the buying decision and/or actually use the product. Yet in almost all product development projects, rarely is only one voice to be heard; more than one group of people has to be considered, although not every group needs to be considered to the same degree. We refer to all groups with a stake in the development, existence, and obsolescence of a product as "stakeholders." For example, stakeholders in the development of a bedside heart monitor used in hospitals would include the various groups within the company producing the product (e.g., manufacturing, design, shipping, purchasing), the administrators within the hospital who are concerned with how much the

[2] See Wheelwright and Clark, *Revolutionizing Product Development,* Chapter 8.

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goods cost, doctors and nurses who deploy the device, patients who will be hooked up to it, and service people who will have to fix it if there is a problem. And even this is not a complete list. Hospital maintenance people who must dispose of the cardboard boxes in which the units are shipped have a stake in the existence of the product, as do the people who decide what should be done with the monitors once they are replaced by newer technology. Again, the voices of all stakeholders should not be given equal weight, but in the event that their needs conflict, compromises can be deliberately made – and in some cases may be avoided entirely.

What Information Should You Gather from the Stakeholders?

Figuring out what to ask stakeholders once these people have been determined depends primarily on the undertaking. Is it an entirely new business for the corporation, requiring a hefty amount of resources, or is it a slight modification to something already being produced? In the first situation, if the company is unfamiliar with many of the issues surrounding the product (e.g., how the customer behaves, how purchasing decisions are made, how distribution channels operate) it will want to do some broad, qualitative searching before it delves into product-specific issues in its inquiries. The more empathetic the team members are toward the product's stakeholders, the higher the likelihood that the company will introduce a satisfying product, since essentially it will be able to critique the product just as eventual customers would. Such ability to understand what pleases and displeases stakeholders will be particularly useful during the design process, for almost certainly design tradeoffs will needed. If the team *truly* understands the context and manner in which the product is to be distributed, sold, used, and disposed of, it will be able to make better decisions and knowledgeable choices among options.

Once this empathic portion of the investigation has been conducted, the team can ask more pointed questions about the particulars of the product to be developed. In all parts of the investigation, the team should keep in mind *when* the product is going to be introduced – i.e., the lag time between assessing stakeholder needs and actual product introduction. There may be important issues at product introduction that are not understood or uncovered at the time of the inquiry. These must be addressed at some point during the overall exploration of customer needs.

Sometimes a product is new not only to the company but to the marketplace. When Apple introduced the iPhone, for example, no comparable offering was available to the general population. When this is the case, those interviewing stakeholders may have to explain their questions carefully or frame them in various ways, since respondents may not be able to visualize or verbalize what they want. For example, if a team wants to develop an entirely new recreational, mechanical transportation device that is not a bicycle, it may have to ask questions about lifestyle, environment, storage space, and mechanical dexterity without ever addressing product-related features, since many respondents might not be able to imagine anything other than a bicycle.

The most effective interactions with stakeholders are through direct observation and personal interviews and ideally should be conducted in the setting in which the product will be used. Quite often it is difficult to gain a real sense of what respondents want or need unless the context in which they will be using the product is appreciated. A cardiac nurse might talk about features of heart monitor alarms that should go off when a patient exhibits arrhythmia. Unless the product developer understands the noisiness of the hospital unit in which the monitor may be used, however, she may not make the right decisions about what the alarms should sound like and how loud they should be. Of course, if the product is considered an enhancement or if the application is fairly well understood, less effort is necessary in conducting extensive qualitative investigations or doing so in the actual setting of use. In general, determining what particular form of interaction to employ will depend on the nature of the questions, the accessibility of the desired respondents, and the time and resources available to the team.

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What Do You Do with the Information You Gathered?

Ideally, the data-gathering stage should produce unedited statements from the respondents, general impressions from the interviewers, and quantitative results from surveys, without interpretation or bias. An excerpt of what a team developing a sleeping bag might have to work could include the following:

> I've had my sleeping bag for 24 years and it hasn't let me down yet. I keep it in my car for those times I'm on the road and just feel like goin' and goin'. Where I end up, I sleep! (40 year old male, hiked the Appalachian Trail 2 times, works as a cook in a restaurant to support his climbing and hiking activities.)
>
> I know I'm supposed to take good care of my sleeping bag, but that doesn't make me do it. The thing stays stuffed in its little bag when its not in use. (20 year old college female; uses bag indoors when there aren't enough beds, and about 1 week per summer camping in the West.)
>
> The users we're looking at are simple, unpretentious, and not too much into image when it comes to sleeping bags. The label on the product only means something if it indicates reliability and quality. They're much more concerned with performance (weight, warmth). (Thoughts of the interviewer.)

Working together, the team should go through the list responses, removing statements that do not make sense, clarifying those that have more than one possible interpretation, separating remarks with more than one message into different points, and eliminating duplicate sentiments. Once this has been done, those touching on the same ideas can be grouped and assigned a single summarizing statement. This summarizing should be done repeatedly until the highest or most abstract level is a more manageable group of about 30 user needs that can be consulted throughout the rest of the process. The original statements should not be discarded, however; if a question about what the user groups really wanted subsequently arises, the information can be reviewed for clarification.

In an ideal world, a product would meet the needs of all stakeholders at an acceptable price. In reality, however, technological, logistical, and financial constraints usually demand that some tradeoffs and compromises be made in the design process. Thus to decide which needs are critical and which are less important, their relative and absolute weights have to be determined.

The *relative* weight can be assigned by having users rank order a list of the summary needs. The *absolute* value of each need can be assessed through a technique called the Kano[3] method. By asking and analyzing questions about the user needs in a particular way (see **Appendix A** for more details), each need can be assigned a label – must have, linear satisfier, delighter, and indifferent, which are defined as follows:

Must Have – no matter how well the product meets the need, the customer simply accepts it as something expected. However, if the need is not met, the customer is very dissatisfied.

Linear Satisfier – the better the product is at meeting the need, the better the customer likes it.

[3] Named for its creator, Professor Noriaki Kano.

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Delighter – if the product fails to include this, the customer is not unhappy. But while its absence does not cause dissatisfaction, its presence will increase customer satisfaction.

Indifferent – the need produces neither satisfaction nor dissatisfaction in the consumer, regardless of whether or not it is met in the product.

Once a label has been assigned to each need, subsequent design tradeoff decisions can be made more intelligently. For example, the innovation team designing a new all-season tent for serious hikers will encounter a conflict between weight and durability. A sturdier tent able to withstand all types of wind, snow, and rain conditions will come at the cost of higher weight (not to mention more expensive material costs). In deciding how much durability to build into the product, the team would be helped by knowing that the user need "the tent will withstand any weather that I may encounter" is considered a "must have" by 80% of respondents, while "the tent must not be too heavy" is a "must have" in the eyes of about 40%. Obviously, "too heavy" has to be clarified, but the point is that the "must have" condition should be met given its importance to such a high percentage of respondents.

Concept Generation

Translating Customer Needs into Product Features

Once the information has been prioritized, one more step is necessary: translating the customer's vocabulary into a form that engineers and others can use. For example, what does it mean to a designer when a customer says, "I want to be able to totally stuff my backpack and throw it around without having it open up on me?" How will the designer know that she has satisfied this need? Some sort of metric must be set up with a corresponding target value so the development team can continually evaluate whether the product it is developing is meeting the user needs. (Note: when choosing the target values, it is useful to benchmark existing products – those within the company and from competitors – to determine what the new product should provide.) Most likely, preliminary specifications will have to be modified as prototypes are built and the feasibility and desirability of the chosen values are evaluated.

In the backpack example, the metric might be that the pack should be able to withstand a certain amount of force while stuffed to the very limits of its capacity. It is important at this point that *solutions* to the needs *not* be incorporated into the product specifications. For instance, those who have previously used packs with zippers might translate the above customer need into a specification that the zippers must be of a certain strength. But zippers are only one way of closing the bag and therefore should not yet be assumed to be part of the design.

What Is a Concept?

Until this point, the team's focus should be strictly upon the questions of "who" and "what": Who are the critical stakeholders? What do the stakeholders want? What should the product offer to meet those demands? The question of "how" should have been avoided so that the design options would not be prematurely limited. Once the "what" questions have been answered, however, solutions can be evaluated, primarily through the creation of product concepts.

The distinctive combination of the answers for the "who," "what," and "how" questions is a product concept's positioning statement, where:

Positioning = Differentiation (what + how) + Segmentation (for whom).

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In addition to the verbal positioning statement, a product concept should illustrate, through the use of text, drawings, and models, the product's performance and technical features; how it should be configured or architected, how it should serve customers, who those customers are, and its meaning to those customers (e.g., status, personality, feel, image).[4] Product concepts may be described in objective measures, such as speed, size, horsepower, and the like; in emotional, subjective terms suggesting how the product should make customers feel; or as a combination of emotional and measurable criteria. The purpose of the product concept is to give the entire development team a common set of objectives upon which to act. Since members of even the most interactive team will make a fair number of decisions on their own, they all must have the same goals in mind if their individual choices are to support each other.

One illustration of a product concept is that created by a group of Harvard Business School students for a class project. The students were working with a large electronics company to develop a product that would be introduced to the "home communicator market" within a two-year time frame. Their extremely vague charter was to create a product to address all the communication needs of a household. Prior to and during the concept development process, the team variously assessed stakeholder needs. They held group brainstorming sessions, did library research, kept home diaries of what unfilled needs they experienced themselves, and conducted extensive personal interviews with target customers. They also researched the technology that either was available at the time or would become so within the following two years.

Joining what they learned about customer needs and possible technologies to what they knew about the company's strategic plans, the team came up with a concept called "The Butler." The Butler was intended to present the image of a device that essentially would "take care of many repetitive and mundane tasks, and help organize and manage the home environment."[5] The concept also included details about the physical product (e.g., that it would include a base station and a portable unit arranged in a client-server relationship, and that there would be connections from the base unit to the electrical system, phone lines, and home appliances to allow for certain types of linkage). With this combination of the butler image and specific product features the team was able to move forward in the development process, using the same "script" to guide them in their decisions.

Concepts come from innumerable sources. Broadly, they can stem from an individual's inspirations or represent the output of a group assembled specifically to generate concepts. By and large, the benefit of the former source is that underlying the ideas of individuals are degrees of creativity and differentiation which are often suppressed in a group. The drawback is that the perspective of one person may not take into account all the important downstream issues surrounding the creation and distribution of a product. The more methodical group approach addresses this downstream problem but often at the expense of product uniqueness and distinction; reaching consensus can frequently arise only after the concept has been so chipped away it represents a bland compromise. If the product's *integrity* is kept in mind, this erosion may be prevented, however.

Product integrity refers to "the extent to which the totality of a product achieves the balance [of numerous product characteristics, including basic functions, aesthetics, semantics, reliability, and economy] and attracts and satisfies customers."[6] Integrity is a result of "internal" elements, such as how well the components of the system fit together or how efficient is its design, and "external" features, such as how well the product and its delivery system fit the needs of the customer. A

[4] See Clark and Fujimoto, *Product Development Performance*.
[5] M, Bauer, T. Corn, S. Mertens, and M. Okuno, *The Butler: Contract Book* (written for Professor Marco Iansiti's "Managing Product Development" course, Harvard Business School, Spring 1994).
[6] Ibid.

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weight machine available in many fitness centers, for example, may appear on the outside to be smooth and efficient, with its shiny metal, clean lines, and space-age appearance. If, however, the pieces underneath the glossy veneer have a bit of play in them, allowing them to slip slightly, the chains may start to rattle and the weight plates might begin to rub against each other. The result will be a clanging, friction-filled mess that does not support the smooth image presented by the exterior. The lack of internal integrity will have affected the external integrity as well, since some of the customer's needs – say, quiet workout, consistent measurement of amount of force exerted by user – remain unfulfilled.

Product Architecture

An important aspect of the product concept affecting both internal and external product integrity is that of "product architecture," which refers to the physical building blocks or chunks assembled to create the whole unit. At one extreme, a product can be created as one entire block with no logical division except at the individual component level. Early mainframe computers employed this model. Every new product was basically an entirely new unit, and the company's customers essentially had to throw out the old one and buy the new. Development lead times were typically long, since the whole product was totally redesigned and reconfigured. Alternatively, the use of modules or separable chunks fit together with well-defined interfaces has become increasingly more popular in product design. The benefits of such a construct are many: parts of the product can be upgraded or fixed easily without affecting the rest of the system; new versions can be introduced to the market frequently; and often the risk of encountering technological problems when bringing out new products can be lessened since the development and debugging can be performed incrementally. Personal computers follow this model. If a user wants to improve his display capabilities or increase his system's memory, he only has to buy a new display unit or replace memory chips, respectively, while keeping everything else the same. The potential drawback of a modular design is that the interfaces can reduce overall product performance. In the case of the personal computer, in order to allow a unit to work with various operating systems (modularizing the operating system), the processor's full capability may not be utilized since it has to be general enough to run with various vendors' products. However, the typical PC user is willing to accept the drawbacks in order to gain the price and upgradability advantages the modular system provides.

Factors that weigh heavily in determining how modular to make a product include: how often the product needs to be changed (e.g., upgrades, add-ons, wear-and-tear replacements), how many different varieties of the same basic unit are desired, what level of performance the system must achieve (flexibility sometimes comes at the expense of performance should the pieces not fit together as perfectly as they could be when designed as an integral unit), and manufacturability and design (is it easier to assign design responsibilities by module or to purchase an entire module from a third party?).

Concept Selection

Once generated, concepts need to be evaluated and pared down. Their evaluation might lead to the discovery of more concepts (either by coming up with a new one while examining a previous one, or by combining already proposed ideas). The list will likely go through streamlining, slight expansion, then more narrowing, until a handful of "finalists" has been selected. There are no hard and fast rules for how the final concept should be selected, but the intent should be to devise the best *overall* solution given the relative and absolute importance ratings of stakeholder needs and company exigencies. Succinctly, the final concept should be checked for internal and external product integrity. Because some ultimate measures of the product's success may not be quantifiable, the qualitative nature of this screening process should be taken seriously. The benefits of having spent enough time up front to truly understand stakeholder needs in a general sense are very

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apparent at this point. Such understanding will help the team choose among the multiple desirable concepts to find the real winner.

Product Engineering and Design

Unhappily, a good concept alone does not ensure success. A good concept will only result in a good product if it is effectively translated into a detailed design with coherence and integrity. Countless products have demonstrated that good design can have a tremendous impact on the success of a product. The Apple iPhone, whose sales greatly exceeded expectations, owed much of its appeal to the functionality and integrity of its design, from the user interface to the overall design. Oxo, a manufacturer of kitchen utensils, owed its rapid growth to its revolutionary handle design. Its "big grip" products were able to earn a premium in a previously price-sensitive market. The field of industrial design has played a significant role in these success stories by emphasizing customer focus, creativity, integrity, and attention to detail.

To many people, engineering refers to the "guts" or insides of a product while design applies to that which is visible. While it is true that designers tend to have more of an impact on the appearance of a product, and the engineers might be more involved in the configuration of its parts, the lines of differentiation are not nearly so precise. Designers at an automobile company, for example, will be involved in creating both the exterior shape of the car and the cooling system hidden beneath it. Similarly, the engineers at a camera company might spend most of their time looking at the film winding mechanism in the camera, but they would also be involved in making decisions about parts that interface with the product users. In all situations, engineers and designers have the goal of determining how to match the needs, or design parameters, provided by the internal customers (e.g., manufacturing, shipping, service) to those of the external customers that were uncovered through the customer needs activities described earlier.

Prototyping

Prototypes are representations of all or a portion of a product; they are created during the development process to accommodate any one of a handful of purposes. The completeness of the prototype varies according to how far along in the process it is created and what purpose it intends to serve. Typically, prototypes are used to meet one or more of the following goals:

1) *Test the feasibility of a concept* – this can be done either through physical or analytical (computer, mathematical) modeling.

   For example: Do the pieces fit together well? Can they be built that way at a low enough cost? If built as designed, will the motor make too much noise or is it properly insulated?

2) *Collect customer/stakeholder feedback on part or all of the concept*.

   For example: Does the unit feel comfortable in the user's hand? Is it sturdy enough that when packed sufficiently, it will arrive at the retailer unbroken?

3) *Communicate an idea (to other members of the company, suppliers, distributors, and so forth)*.

   For example: When trying to get management to invest more money so that a particular concept can be pursued rather than the cheaper one, it is helpful to show the differences between the two.

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Knowing why a prototype is being created is an important first step toward having it provide its intended benefits. In addition to identifying the purpose, however, questions about how accurate and complete the prototype should be need to be answered. A small appliance manufacturer wanted to make sure, before it ordered molds, that the newly designed blender would not overheat when run through reliability testing. To avoid incurring the expense of having just one molded unit built, engineers used an existing unit with the same dimensions but of a slightly different shape to run the tests. The motor did heat up but was about 5% below the allowed maximum, and the molds were ordered as a result. When actual tests were run on the first blenders made from production parts, however, the units heated up to 5% *above* the maximum. The team had made the mistake of not realizing how accurate the prototypes needed to be or what range of results would be acceptable given the slightly different parts it was using.

Also at issue in the development of prototypes is who should be involved in their construction and testing. Quite often, the functional area directly affected by the results performs the tests. In the blender case, the most salient issue being investigated was technical performance, and therefore the engineers built the prototypes and performed initial tests. Later in the process, as aesthetic features became more important, marketing and industrial designers did prototype work. In this arrangement, little information about one series of tests spread throughout the organization so that everyone could know what was being learned. As a result, subsequent tests or decisions made by members of the development team not present for the building and testing done earlier failed to incorporate the information already gathered.

Pilot Production, Ramp Up, and Market Launch

As suggested in the example above, prototypes are often used throughout the development process to prove the manufacturability of a particular design. If prototyping has been effective, therefore, by the time the final design has been accepted the factory should be nearly ready to execute full-scale production. This final move toward commercial production entails three steps: pilot production, ramp up, and market launch.

Pilot production is the stage at which a factory first uses parts produced by the molds, machines, and suppliers that will ultimately be employed when the products are being shipped to customers. The purpose is to test the notion that the company is ready to ship the product to customers, and the goal is to confirm the assembly, test, and other operational procedures used in full-scale production. All parts suppliers should be ready to support anticipated levels of production, and all equipment should be in place in the factory.

Once the basic process has been established, the manufacturing facility needs to develop the capability to produce the volume of units that are expected within the first year or so. The phase is termed "ramp up" since the increase is generally gradual, rather than in the form of a step function. This should be a learning and adjusting period for all those concerned (manufacturing, design, suppliers, shipping, and so forth) since increased volumes typically reveal problems that have not appeared previously.

Market launch is the phase in which the marketing plans regarding the target segment, positioning, product and service combinations, pricing, marketing communications, and distribution are executed in an integrated way to ensure the right product gets to the right customers at the right time for the right price.

At this point, the success of the development process can begin to be gauged. The Stage-Gate Model calls this the Post-Implementation Review. If the process was performed properly, manufacturing ramp up should be short and lacking major surprises (although inevitably there are some problems), the product should begin to flow through the distribution channels, and the

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anticipated demand should become visible. As with any other experience, however, the procedure should not be considered complete until an analysis of what went well and not-so-well during the development process has been performed and the relevant lessons taught throughout the organization.

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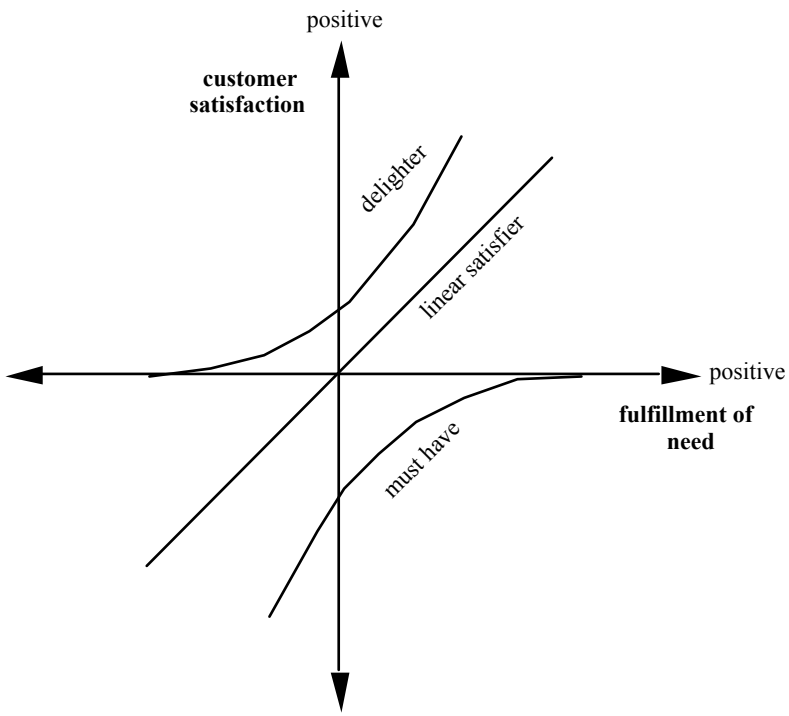
Appendix A: The Kano Method

The Kano Method recognizes that the degree of fulfillment of a customer requirement does not always correlate in one predictable manner with the level of customer satisfaction. In other words, the relationship between fulfillment and satisfaction is not always the same nor is it always linear. For instance, every increase in horsepower for a new car might not give the customer the same increase in happiness. Moving from 100 to 150 horsepower might be a big improvement in the customer's eyes, but a move from 300 to 350 may not be quite so important if 300 hp were more than enough for the target customer base. This is an example of a non-linear attribute. **Figure A1** displays qualitatively a few of the different relationships that can exist between the fulfillment of a need and the degree of customer satisfaction. A "delighter" is one that can only have positive customer satisfaction; its absence does not lead to customer dissatisfaction (i.e., a point below the x-axis). On the other hand, a "must have" will never make a customer happy. It is expected that the product have the feature, so the best that a development team can hope to do is find out how much of the feature (how far to the right on the x-axis) the product should have so the customer will not be dissatisfied. (**Figure A1**, a qualitative visual tool, is not meant to indicate any precise or measurable relationships.)

Figure A1: The Kano Method



The Kano method is frequently implemented through a customer questionnaire. Two questions, one positive and one negative, are included for each customer need. For example, if a user need for a new mountain bike were that the rear tire should be easy to change, the positive question would read: "If the rear tire is easy to change, how do you feel?" while the negative one would ask, "If the rear tire is not easy to change, how do you feel?" For each question, the respondent is given five options: "I like it that way," "It must be that way," "I am neutral," "I can live with it that way," "I dislike it that way." The responses are processed by using a "two-dimensional table of evaluation" (see **Figure A2**) to translate them into the different labels.

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Figure A2: Kano Method Evaluation Table

Result of Positive Question	Result of Negative Question: Like	Must Be	Neutral	Live with	Dislike
Like	Q	D	D	D	L
Must Be	R	I	I	I	M
Neutral	R	I	I	I	M
Live With	R	I	I	I	M
Dislike	R	R	R	R	Q

I = Indifferent
R = Reverse
Q = Questionable Result
M = Must Have
L = Linear Satisfier
D = Delighter

Adapted from G. Burchill. *Concept Engineering: An Investigation of TIME vs. MARKET Orientation in Product Concept Development*, 1993.

The labels respondents assign must be summarized for each need before they can be used in the design process. Ideally, one summary label would be attached to each need based on the responses. For example, if one feeling predominates (e.g., 70% of respondents consider a particular need to be "must have"), that label can be assigned to the need. If, however, there is less harmony among the various respondents (e.g. 40% see it as "must have," 35% as "linear satisfier," 15% as "indifferent," and 10% as "reverse"), assigning one label would misrepresent the level of agreement between customers. In this case, instead of simply keeping one label attached to the user need, it might be better if the percentages for each label were recorded for those particular needs.

Once summarized for each need, the labels can be used to prioritize product requirements during the design process. Most development projects involve tradeoffs among product requirements. If, for example, both "make the rear tire easy to change" and "make the bike light" were customer requirements, the team might be confused about whether to improve the ease of changing the rear tire should doing so make the bicycle heavier. The decision might be easier if the development team knew that 60% of the respondents thought the lightness of the bike was a "must have" (assuming the definition of "light" has been clarified), while only 20% of the respondents thought the ease of changing the rear tire was a "must have." This information would indicate to the team that it should make the rear tire easier to change only if it did not negatively affect the weight of the bike.

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